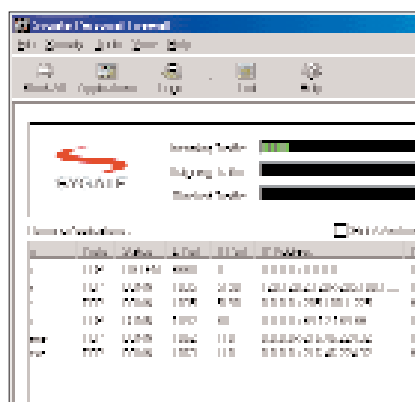


firewalls open up a response box which you can use to allow or disallow that application, and also tell the firewall to remember this rule for future reference. This is a very important feature because it is the best way of knowing if some unknown application (or Trojan) is trying to access the Internet. In some firewalls, application-level filtering was bundled along with the feature to block ports or protocols. Sygate was the best in this feature. After you use an application for the first time and create a rule to allow it, you can specify the IP addresses that this application can use, the range of ports that are allowed, and even schedule a period-range during which the application can be run. ZoneAlarm Pro also offers some features related to application-level filtering in terms of ports and protocols that are related to the application. You can also specify whether the application has access to the local zone or Internet zone, and whether the application can act as a server to other programs trying to connect to it.

Blocking IP addresses and Web sites:

Although it is not traditionally part of a firewall's job, most firewalls nowadays sport this feature. They can be used to block access to malicious Web sites or as a content filtering tool. They can even restrict access to machines in the local network. Most firewalls just ask you to specify the IP addresses that have to be banned. Norton allows you to ban IP addresses or host names (including subnets) in conjunction with the protocol. PGP works in a similar fashion. ZoneAlarm Pro has a separate module that allows you to create a restricted zone based on IP address or



Sygate is the easiest to configure

host name. In some firewalls, you can cut off access to the Internet using the IP address range, or by setting the security level to the highest. ZoneAlarm Pro has a single icon that allows you to immediately stop access to the Internet.

Into the Network Neighbourhood:

Intruders don't just exist in the Internet space—they could be people who are sitting next to you in a cyber cafe or in office and sharing the same network connection. So, firewalls should ideally also protect you from intrusions coming from the local network, and even deny certain computers access to files shared on your PC. Most firewalls didn't sport this feature, and even if they did, it didn't work properly and we could easily access shares on the Test Bench. eSafe, Norton, ZoneAlarm and ZoneAlarm Pro were the only ones that allowed disabling of file sharing to banned computers on the local network. In Norton, you could do that by restricting protocols to local addresses, while ZoneAlarm

and the Pro version have separate settings for the local network. In eSafe, the personal firewall module doesn't allow blocking Network Neighbourhood, but in the Administrator module, under the privileges for the user, you can cut off access to local networks.

Other features: Different firewalls came with many other utilities that added to the feature list. Norton and eSafe have the privacy or protection of confidential information feature. Here, you can specify confidential information that you do not want transmitted to any Web site or non-secure site. So, you can easily enter some digits of your credit card, and block this information from being passed. Norton also allows you to block referrer logging, that is, sites you visit will not know where you came from. You also have the provision to block cookies. Some firewalls also come with password protection features. Tiny is the only one that has a remote administration utility. eSafe also ships an antivirus package along, while PGP has features relating to digital signatures and protection for VPNs (Virtual Private Networks).

Among the various firewalls, it is evident that ZoneAlarm Pro and Norton Personal Firewall 2001 have the maximum number of features.

Ease of use

A firewall may perform well and have loads of features, but if it is difficult to configure and access the features, then all the good qualities are of no use. Ease of use is therefore an important criterion in choosing a firewall, and we looked at factors such as interface, configuration and reporting to judge the firewalls.

Configuration: With most firewalls, you could start working out of the box using just the default setting, without having to configure anything. This is true especially in the case of application-level firewalls. Here, every time an application tries to

Tracing back the Intrusion

If you are facing a number of intrusion detection attempts, then if you have the time and inclination, you can take some proactive measures. Most firewalls tell you the IP address of the machine that made the intrusion attempt. A firewall such as NeoWatch also comes with the NeoTrace utility that allows you to backtrace to where the intrusion attempt came from, and also report the intrusion to its Web site. NeoTrace basically performs a trace route operation on the IP address to find out where the address originated. You can do the same using the command `tracert <IP address>` in your DOS prompt. ZoneAlarm also allows you to report an intrusion to

Zone Labs, and also monitors the information that is sent, so you can even keep your identity a secret. You should report such attacks as it helps security organisations build a database of places from where intrusion attacks are being made, and try to ensure that they don't happen again. If you get the IP address, even you can report the intrusion to the ISP from where the attack came or to the domain-hosting provider. The ISPs and Domain space providers normally have a policy wherein their IP addresses cannot be used for such intrusion activities, and can therefore remove the offender from their network.

[illegible]

Filter the results in NeoWatch